PulseMargin

Return-risk scoring at checkout, inside Exasol

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The problem

Indian e-commerce bleeds margin to returns:

- **COD refusals** — the parcel is rejected at the door. Seller pays two-way shipping.
- **Size bracketing** — “order M and L, return one.” Apparel return rates exceed 30% in high-RTO pincodes.

Today this is caught after fulfilment, when the only options are a refund and a loss.

The decision is at checkout

Is **this cart**, from **this customer**, to **this pincode**, on **this payment mode**, likely to come back?

Answer it in one query, before the parcel ships. Then intervene:

- Verify sizing before dispatch
- Convert COD to prepaid
- Route to a cheaper courier tier

What we built

A Go service that takes a cart and returns an interception tier, scored entirely **in-database**.

- No model server. No Python microservice. No feature store to keep in sync.
- Feature aggregation + logistic regression run as a `PYTHON3` UDF inside Exasol.
- One round trip. Latency is a property of the warehouse.

How it works

1. Checkout service POSTs the cart to `/api/v1/checkout/score`
2. One CTE query joins: customer history · pincode RTO · SKU sizing skew
3. `predict_return_risk` UDF scores: customer return rate, pincode RTO, size bracketing, size skew, cart value, COD
4. Result maps to `LOW` / `MEDIUM` / `HIGH_RISK` with tailored interventions

Why Exasol

- **In-database scoring** — the join across 1M customer profiles, pincode and SKU dimensions is what the warehouse is built for.
- **Elastic, parallel UDFs** — Python runs alongside the data, not over the network.
- **Single source of truth** — risk features and the model move together with the data.

Demo

```
curl -s localhost:8080/api/v1/checkout/score \
  -H 'Content-Type: application/json' \
  -d '{"checkout_id":"chk_101","customer_id":"CUST_0000042",
       "pincode":"110001","payment_mode":"COD",
       "items":[{"sku":"SKU-0001","category":"Apparel_Jeans","size":"M","price":1899},
{"sku":"SKU-0001","category":"Apparel_Jeans","size":"L","price":1899}]}'
```

→ risk_score 0.95 · HIGH_RISK · 102 ms in Exasol Two sizes of one SKU (bracketing) + COD + a high-RTO pincode floors the cart.

Status

- Working end to end on Exasol Personal: schema, UDF, 1M-row seed, live API, web UI.
- Go · `go test` green · CI on GitHub Actions with the toolchain pinned via `mise`.
- MIT licensed. Minimal Oat UI embedded in the binary.

github.com/tachyons/pulsemargin